to move either office, you come up with a natural suggestion: Build a bridge from one office to the other, connecting the two towers. This allows employees to move freely between the offices without having to go down and then up ninety floors.

A wormhole plays a similar role: It is a bridge or tunnel that provides a shortcut from one region of the universe to another. Using a two-dimensional model, imagine that a universe is shaped as in Figure 11.1.

If your corporate headquarters are located near the lower circle in 11.1 (a), you can get to your field office, located near the upper circle, only by traversing the entire U-shaped path, taking you from one end of the universe to another. But if the fabric of space can tear, developing punctures as in 11.1(b), and if these punctures can "grow" tentacles that merge together as in 11.1(c), a spatial bridge would connect the previously remote regions. This is a wormhole. You should note that the wormhole has some similarity to the World Trade Center bridge, but there is one essential difference: The World Trade Center bridge would traverse a region of existing space—the space between the two towers. On the contrary, the wormhole creates a new region of space, since the curved two-dimensional space in Figure 11.1(a) is all there is (in the setting of our two-dimensional analogy). Regions lying off of the membrane merely reflect the inadequacy of the illustration, which depicts the U-shaped universe as if it were an object within our higher-dimensional universe. The wormhole creates new space and therefore blazes new spatial territory

Do wormholes exist in the universe? No one knows. And if they do, it is far from clear whether they would take on only a microscopic form or if they could span vast regions of the universe (as in Deep Space Nine). But one essential element in assessing whether they are fact or fiction is determining whether or not the fabric of space can tear.

Black holes provide another compelling example in which the fabric of space is stretched to its limits. In Figure 3.7, we saw that the enormous gravitational field of a black hole results in such extreme curvature that the fabric of space appears to be pinched or punctured at the black hole's center. Unlike in the case of wormholes, there is strong experimental evidence supporting the existence of black holes,